

Roll No.....

I BE CLASS TEST-III May 2024
Electronics & Telecommunication
1AMRC1 Applied Mathematics-I

Kavyansh

Time-70 mins.

Maximum Marks [20]

Note-Attempt all the questions.

Q.1 If $u = x^2 - 2y^2, v = 2x^2 - y^2$ where $x = r \cos \theta, y = r \sin \theta$,
show that- [4]

$$\frac{\partial(u, v)}{\partial(r, \theta)} = 6r^3 \sin 2\theta.$$

Q.2 Find the area of the loop of the curve- [4]

$$x(x^2 + y^2) = a(x^2 - y^2).$$

Q.3 Find the entire length of the cardioid- [4]

$$r = a(1 + \cos \theta)$$

and show that upper half is bisected by the line $\theta = \frac{\pi}{3}$.

Q.4 Evaluate-

$$\iint_R xy dx dy,$$

where R lies in the circle $x^2 + y^2 = a^2, x \geq 0, y \geq 0$. [4]

Q.5 Evaluate

$$\int_0^a \int_y^a \frac{x}{x^2 + y^2} dx dy,$$

by changing the order of the integration. [4]

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